

## The state of US health, 1990-2010: burden of diseases, injuries, and risk factors.

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**IMPORTANCE:** Understanding the major health problems in the United States and how they are changing over time is critical for informing national health policy. **OBJECTIVES:** To measure the burden of diseases, injuries, and leading risk factors in the United States from 1990 to 2010 and to compare these measurements with those of the 34 countries in the Organisation for Economic Co-operation and Development (OECD) countries. **DESIGN:** We used the systematic analysis of descriptive epidemiology of 291 diseases and injuries, 1160 sequelae of these diseases and injuries, and 67 risk factors or clusters of risk factors from 1990 to 2010 for 187 countries developed for the Global Burden of Disease 2010 Study to describe the health status of the United States and to compare US health outcomes with those of 34 OECD countries. Years of life lost due to premature mortality (YLLs) were computed by multiplying the number of deaths at each age by a reference life expectancy at that age. Years lived with disability (YLDs) were calculated by multiplying prevalence (based on systematic reviews) by the disability weight (based on population-based surveys) for each sequela; disability in this study refers to any short- or long-term loss of health. Disability-adjusted life-years (DALYs) were estimated as the sum of YLDs and YLLs. Deaths and DALYs related to risk factors were based on systematic reviews and meta-analyses of exposure data and relative risks for risk-outcome pairs. Healthy life expectancy (HALE) was used to summarize overall population health, accounting for both length of life and levels of ill health experienced at different ages. **RESULTS:** US life expectancy for both sexes combined increased from 75.2 years in 1990 to 78.2 years in 2010; during the same period, HALE increased from 65.8 years to 68.1 years. The diseases and injuries with the largest number of YLLs in 2010 were ischemic heart disease, lung cancer, stroke, chronic obstructive pulmonary disease, and road injury. Age-standardized YLL rates increased for Alzheimer disease, drug use disorders, chronic kidney disease, kidney cancer, and falls. The diseases with the largest number of YLDs in 2010 were low back pain, major depressive disorder, other musculoskeletal disorders, neck pain, and anxiety disorders. As the US population has aged, YLDs have comprised a larger share of DALYs than have YLLs. The leading risk factors related to DALYs were dietary risks, tobacco smoking, high body mass index, high blood pressure, high fasting plasma glucose, physical inactivity, and alcohol use. Among 34 OECD countries between 1990 and 2010, the US rank for the age-standardized death rate changed from 18th to 27th, for the age-standardized YLL rate from 23rd to 28th, for the age-standardized YLD rate from 5th to 6th, for life expectancy at birth from 20th to 27th, and for HALE from 14th to 26th. **CONCLUSIONS AND RELEVANCE:** From 1990 to 2010, the United States made substantial progress in improving health. Life expectancy at birth and HALE increased, all-cause death rates at all ages decreased, and age-specific rates of years lived with disability remained stable. However, morbidity and chronic disability now account for nearly half of the US health burden, and improvements in population health in the United States have not kept pace with advances in population health in other wealthy nations.